

Hive-Nav: Belief-Guided Multi-Robot Room Classification and Open-Vocabulary Object Search

Abstract—Open-vocabulary multi-robot target search in buildings involves two coupled tasks: identifying relevant rooms and locating a queried object. Real-world deployments expose challenges often under-represented in simulations. Visually similar rooms may provide little discriminative evidence, while offloading concurrent inference requests from multiple robots to a shared GPU over lossy Wi-Fi introduces variable latency. We present Hive-Nav, a two-phase multi-robot framework. Phase 1 fuses VLM-derived signed evidence across frames and robots into a shared room-level belief map. Phase 2 combines this map with a semantic room-type hint to assign candidate rooms to robots, which then perform coverage-based searches to detect and approach the target. We selected Qwen3-VL-4B after benchmarking 11 open-source VLMs. It achieved 94.5% curated-room accuracy using 3.3 GB of VRAM. We evaluated Hive-Nav using a three-robot team in a six-room academic building and through controlled Gazebo simulations. Across 10 physical Phase 1 trials, our proposed multi-visit fusion framework increased room-classification accuracy from 88.3% for the single-robot, single-visit baseline to 98.3% with the three-robot team. All 15 physical Phase-2 search-and-approach trials succeeded using the stored semantic map. In simulation, three robots with shared memory reduced Phase 2 makespan by 19.6% and travel by 32.3% relative to one robot. Without shared memory, three robots were only 0.9% faster despite performing 3.1 times more room-discovery work, indicating that shared semantic memory, rather than team size alone, drives search efficiency under constrained inference.

I. INTRODUCTION

A mobile robot that receives a task such as “find an oven in the kitchen,” must solve two problems: a) it must correctly classify a room as a kitchen, and b) it must locate the oven within the room. When a team of robots shares this task, the first problem becomes collaborative as each robot can contribute observations of the room it visits, and the resulting semantic map should guide the second problem by narrowing down the search to the most probable rooms. Recent multi-robot navigation systems have made rapid progress in this pipeline, combining classical frontier-based exploration and coordination [1], [2] with Vision-Language Model (VLM) reasoning to guide a team of agents toward open-vocabulary object goals [3]–[5].

However, real-world deployment reveals a critical perception failure that simulation does not expose. For instance, our target environment in this work, an academic building with six rooms (four labs and two office areas), shared a highly similar generic visual vocabulary (benches, monitors, equipment racks) and differed mostly in object arrangement. During initial deployment, YOLO-World [6] detected zero discriminative objects in four of the six rooms, making open-vocabulary room classification impossible. Direct Vision-Language Model (VLM) room labeling was an apparent alternative, but multimodal language models can hallucinate visual content [7] and remain sensitive to perspective and viewpoint changes [8], degrading downstream navigation and affordance prediction [9], [10]. In pilot deployments, a single frame of a cluttered lab sometimes produced a confident but incorrect label, and nearby frames repeated the same

error. Therefore, a reliable room classifier must accumulate multi-view evidence and calibrate confidence to ensure that coordination decisions are driven by evidence rather than heuristic thresholds.

Hardware deployments introduce another challenge that is typically absent in simulations. Real-time VLM inference on resource-constrained robots typically requires off-board processing [11], [12]. Consequently, multiple robots must contend for a shared central GPU over Wi-Fi. Prior work has examined multi-robot computation offloading and inference scheduling [11], [13], [14]. However, contention among robots performing deadline-constrained semantic room classification remains under-explored. Moreover, deploying open-vocabulary detectors like YOLO-World [6] and Grounding DINO [15] on edge devices introduces latency challenges heavily exacerbated in multi-robot settings [16]. To address these practical challenges, we present Hive-Nav, a multi-robot framework that builds a shared semantic room map to guide open-vocabulary object search in real-world environments. Hive-Nav operates on a precomputed occupancy map, reflecting many structured indoor settings, including offices, hospitals, academic buildings, and warehouses, where the structural layout is known but semantic labels or object locations may be unavailable, incomplete, or change over time [17], [18]. The system maps these unknown semantics and executes target searches using a two-phase pipeline. Phase 1 fuses multi-view VLM object inventories into a normalized per-room belief distribution using a custom evidence table and weighted log-evidence accumulation. Phase 2 consumes this belief map to rank candidate rooms and execute coverage-based search with an open-vocabulary detector with a given target object.

The primary contributions of this study are as follows:

- 1) A two-phase framework that fuses multi-view VLM evidence into shared room-level beliefs and coordinates open-vocabulary target search, validated on a physical three-robot team.
- 2) A disagreement-triggered multi-visit verification protocol that raises real-world room-classification accuracy from 88.3% to 98.3%, with quantified efficiency trade-offs.
- 3) A controlled ablation showing that stored room beliefs prevent repeated classification from eroding multi-robot gains. Without memory, three robots are only 0.9% faster despite performing $3.1\times$ the room-discovery work.
- 4) A comparative evaluation of 11 open-source VLMs across accuracy, reliability, runtime, and memory use, together with real-world Wi-Fi measurements that motivate on-demand frame capture for shared off-board inference.

II. RELATED WORK

Semantic exploration has progressed from single-robot object-goal navigation [19] to methods that embed pretrained vision-language representations in persistent maps [20] and hierarchical open-vocabulary 3D scene graphs [21]. Extending these capabilities to robot teams introduces an additional

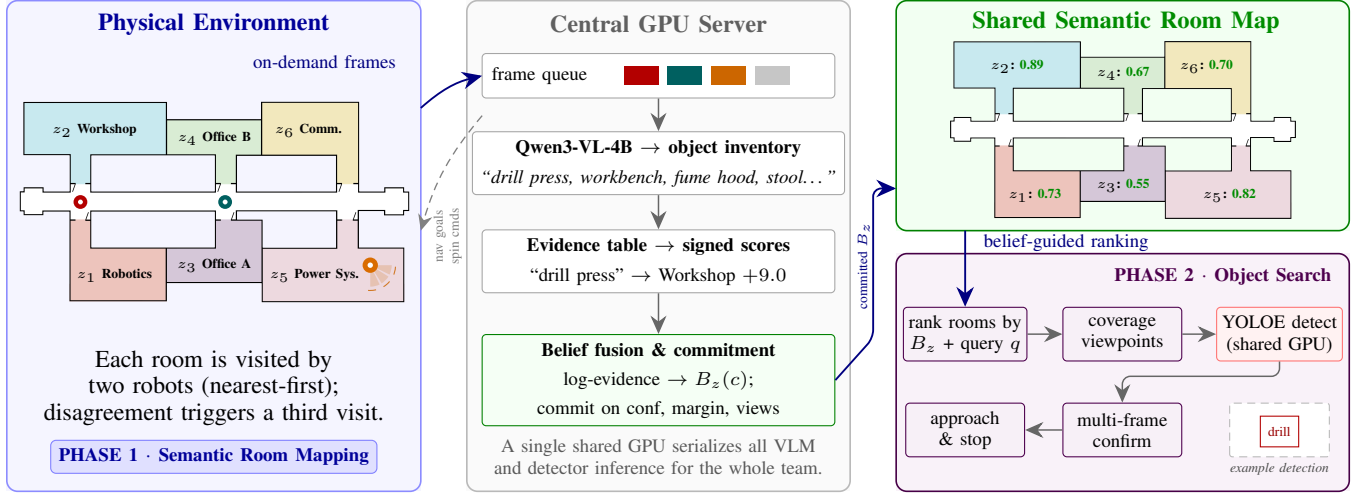


Fig. 1. Overview of **Hive-Nav**. (Left) Robots first visit rooms closest to them. Each room is observed by two robots, with a tie-breaking third visit on disagreement. (Centre) A shared GPU serializes VLM inference, mapping Qwen3-VL object inventories through an evidence table into class scores fused into per-room belief distributions. (Right) The belief map ranks candidate rooms by relevance to the query, then coverage viewpoints are generated and the target is detected and approached.

challenge of efficiently sharing maps and semantic evidence across robots. MCoCoNav [22] uses multimodal chain-of-thought reasoning to compute probabilistic exploration-value scores for candidate frontiers and fuses observations through a shared global semantic map. Co-NavGPT [3] centralizes planning in an LLM that aggregates per-robot sub-maps into a unified map and assigns frontiers to robots. COMRES-VLM [4] combines frontier-cluster extraction, topological skeleton analysis, and VLM reasoning over shared occupancy maps to generate globally consistent waypoint assignments. Like Hive-Nav, it is evaluated on a physical robot team. Each robot runs SLAM Toolbox and Nav2 onboard for local waypoint following, while a central off-board computer performs VLM planning using map and pose updates transmitted over Wi-Fi. It differs from this study in scope, with coordination targeting frontier exploration of unknown space rather than classifying a fixed room set with persistent per-target cross-robot belief fusion. GoalSwarm [5] is the most closely related framework to our work. It offers a fully decentralized multi-UAV framework where each agent builds a shared 2D semantic occupancy map from a detector on a shared external GPU, with map fusion and task coordination kept peer-to-peer rather than centrally planned.

None of these systems adopts a multi-robot classify-then-search architecture or fuses semantic evidence at room level. Instead, they maintain dense shared maps or per-pixel value fields. Hive-Nav addresses this gap through a two-phase architecture that fuses observations from a homogeneous three-robot ground team into compact room-level beliefs and uses this shared semantic memory to guide open-vocabulary search over a precomputed occupancy map.

III. PROBLEM FORMULATION

a) *Environment and mission structure*: Let the robot team be denoted by $\mathcal{R} = \{r_1, \dots, r_R\}$ and the physical rooms in a known occupancy map by $\mathcal{Z} = \{z_1, \dots, z_N\}$. Each room z has an unknown semantic class $y_z \in \mathcal{C}$, where $\mathcal{C} = \{c_1, \dots, c_K\}$ is the set of candidate classes. The mission proceeds in two phases. In Phase 1, each robot greedily selects the nearest unclassified room, where every room is

visited by at least two distinct robots and a third is dispatched if the beliefs disagree. Each visit captures observations and produces per-frame signed evidence scores $s_{i,z,f,c}$ (robot i , room z , frame f , class c) using a VLM inventory and evidence table, fused into a shared categorical belief $B_z(c)$. In Phase 2, given a natural-language target query q , the team uses shared beliefs to rank the candidate rooms and search for the target with an open-vocabulary detector. Hive-Nav uses Qwen3-VL-4B [23] for room-evidence extraction and YOLOE [24] for query-conditioned object detection.

IV. HIVE-NAV SYSTEM OVERVIEW

A. Hardware Platform

Hive-Nav operates on a team of three DJI RoboMaster EP Core ground robots, each equipped with a Raspberry Pi 5, RPLiDAR A2M12 2-D LiDAR, and the RoboMaster’s built-in camera (1280×720). Each Raspberry Pi runs ROS2 Jazzy [25] with Nav2 [26], AMCL localization [27], and a domain-bridged transport layer. VLM and open-vocabulary detector inference, together with belief fusion, are offloaded to a central workstation equipped with an Intel Core Ultra 7 265K processor, 32 GB of RAM, and an NVIDIA GeForce RTX 3050 GPU with 8 GB of memory. To communicate with the off-board computer over the university’s Wi-Fi, the robots utilize ROS2 domain bridging. This architecture isolates each robot’s control loop on a separate DDS domain while cross-publishing key inference topics, such as frame requests, belief updates, and Phase 2 commands, onto a central domain, leveraging ROS2 Quality-of-Service (QoS) policies to mitigate bandwidth constraints.

B. Software Architecture

Hive-Nav’s architecture distributes workloads between the robot edge and a central server. On each Raspberry Pi, the robot-side component manages navigation, localization, and camera control. Rather than streaming continuous video, a room-visit executor captures and transmits JPEG-compressed frames strictly on-demand at designated poses, reducing bandwidth demand and tying each call to a precise viewpoint. These requests are sent to the central computer, which

hosts a mission-coordination node to manage state and room allocation, alongside an inference server. This server serializes all VLM and detector requests through a single GPU lock, utilizing a priority queue to handle Phase 2 approach snapshots.

C. Two-Phase Mission Flow

Figure 1 shows the complete pipeline. The mission proceeds in two sequential phases:

a) Phase 1 - Semantic Room Mapping: This work employs a greedy nearest-neighbor multi-robot task allocation (MRTA) strategy (classified as ST-SR-IA in [28]). Each robot queries its Nav2 planner for shortest-path distances to all unvisited rooms and claims the nearest one. The rooms are prioritized in a fixed order, with unvisited rooms taking precedence over those needing a second observation due to low classifier confidence, which in turn take precedence over rooms requiring a tiebreaker visit due to conflicting labels. To handle suboptimal initial assignments caused by the first-come-first-served nature of the allocation, an auction-style task reassignment mechanism [29] is incorporated, where if a robot measures a path to a claimed but not yet occupied room that is shorter than the current claimant's by a set threshold, it displaces the original assignment, and the displaced robot is reassigned.

Entry waypoints are prespecified, while inside viewpoints are computed at runtime by flood-filling from the room entry into free space and scoring cells by room depth, visibility coverage, and local clearance. Each visit is assigned a budget of 8–9 frames: one entry frame, one to two transit frames, and a 6-pose, 60°-spaced sweep at the inside viewpoint. Every room is independently visited by at least two robots, and a third is dispatched if the first two commit different labels, after which the robots return to their spawn locations.

b) Phase 2 - Object Search and Approach: Using the shared belief map, each robot navigates to coverage-based viewpoints in its assigned candidate rooms, captures on-demand snapshots during small in-place yaw sweeps and while traveling, and waits for the open-vocabulary detector (YOLOE) to confirm the target before approaching it with a stop-look-move controller that stops when the target's bounding box fills a prescribed fraction of the frame. Full details are given in Section VI.

V. PHASE 1: SHARED SEMANTIC ROOM MAPPING

A. Evidence-Based Room Scoring

Rather than prompting the VLM to directly output a room label (which risks hallucination and overconfidence on ambiguous frames [8]), Hive-Nav prompts the model to produce an inventory of visible objects and room features. Each frame captured during a room visit is sent to a 4-bit quantized Qwen3-VL-4B on the central computer, instructed to list discriminative items without inferring the room's class.

A hand-crafted evidence table maps inventory phrases to signed scores across the six room classes: 57 phrases in total, 4 to 19 per room. The phrases and scores were derived exclusively from separate pilot deployment logs and frozen before the Phase 1 evaluation; no evaluation trials were used to revise them. Exclusive/unambiguous items (e.g., “laser cutter”, “yellow electrical box”) receive weights of 8-10, while shared furniture appearing across several rooms (e.g.,

“bench”, “office chair”) receives low weights of 1-2. A few visually competing room pairs carry a small negative cross-penalty so one room's confirmed sighting cannot be diluted by another's generic furniture. Case-insensitive substring matching between the VLM inventory and the table adds a phrase's signed weights to a per-frame vector $s_{izf} = (s_{izf1}, \dots, s_{izfK})$, where $K = 6$. Figure 2 shows this pipeline on a real frame from the Workshop Lab. The frame contributes +9.5 to Workshop, +5.0 to Communications, and +3.5 to Power Systems, demonstrating why room identity should not be inferred from a single frame.

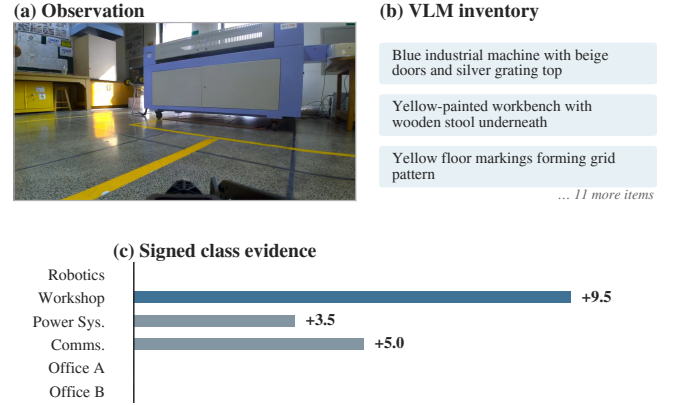


Fig. 2. Qualitative example of Hive-Nav's evidence-scoring pipeline on a single frame from the Workshop Lab (Room 2). (a) The observation contains equipment shared with several room classes. (b) The VLM generates an open-vocabulary object inventory (excerpt shown). (c) The evidence table converts matched inventory phrases from the complete inventory into signed scores over the six room classes.

B. Belief Fusion and Commitment

a) Within-visit fusion: For robot i visiting room z , each frame's signed evidence vector is centred to produce a zero-mean relative-evidence representation:

$$\tilde{s}_{izfc} = s_{izfc} - \frac{1}{K} \sum_{d=1}^K s_{izfd}.$$

The centred scores are accumulated with per-frame quality weights:

$$E_{izc} = \sum_{f=1}^{F_{iz}} w_{izf} \tilde{s}_{izfc}.$$

where $w_{izf} = w_{\text{sharp}} \cdot w_{\text{yaw}}$. Here, w_{sharp} reduces the weight of blurred frames, and w_{yaw} reduces the weight of a frame landing in a yaw bin already covered. This matters in practice: wheel drift means the six 60° spaced sweep poses do not land exactly 60° apart, and a frame dropped over Wi-Fi is retried while the robot keeps moving. So, the retry can arrive from an already-covered heading. Weight w_{yaw} discounts it rather than letting it double-count as an independent view. The per-visit belief score is a softmax with a uniform prior $\pi_c = 1/K$:

$$b_{iz}(c) = \frac{\pi_c \exp(\beta E_{izc})}{\sum_{d=1}^K \pi_d \exp(\beta E_{izd})}.$$

Because the evidence-table weights are hand-designed scores rather than calibrated likelihood ratios, b_{iz} is reported as a normalized belief score rather than a calibrated posterior probability.

b) Cross-robot fusion: Each visit provides an accumulated evidence vector E_{iz} and a quality score q_{iz} equal to the number of distinct yaw bins observed during the visit. Rather than averaging the normalized per-visit beliefs, the coordinator combines the underlying evidence using normalized quality weights, following the log-odds evidence-accumulation principle common to Bayesian occupancy-grid fusion [30]. This evidence-level fusion strategy has been shown to maintain robustness against correlated viewpoint failures in collaborative mapping [31]:

$$\rho_{iz} = \frac{q_{iz}}{\sum_{j \in \mathcal{R}_z} q_{jz}},$$

$$\bar{E}_{zc} = \sum_{i \in \mathcal{R}_z} \rho_{iz} E_{izc},$$

$$B_z(c) = \text{softmax}_c(\log \pi_c + \beta \bar{E}_{zc}),$$

where $\mathcal{R}_z$ contains the robots that observed room z , label disagreement controls whether a third observation is requested, and B_z provides the final shared room belief.

c) Commitment and termination: Phase 1 terminates when all rooms are settled via label agreement or exhausted robot visits. Persistent disagreements are logged as unresolved conflicts rather than false consensus.

VI. PHASE 2: QUERY-DRIVEN OBJECT SEARCH AND APPROACH

Once Phase 1 ends, the robots return to their spawn locations and await a user query. A query consists of a natural-language target description or a reference image along with a room hint. If only a reference image is provided, the VLM generates a text description to prompt the open-vocabulary detector so the image alone is not used at detection time. A room hint restricts the eligible rooms using the Phase 1 labels. Otherwise all rooms remain eligible. Robots repeatedly claim the nearest eligible room according to the Nav2 path distance, while shared claims prevent duplicate searches.

a) Coverage Viewpoint Generation: Within each candidate room, the system generates viewpoints that maximize the planned camera coverage. Starting from a computed seed inside the room, the occupancy grid is flood-filled within a configurable search radius and clipped at the room entrance to restrict corridor leakage. A clearance mask removes cells near obstacles and connected-component filtering excludes the unnavigable area. Candidate positions and camera headings are evaluated using camera range, field-of-view, and map-based line of sight. A greedy selector then chooses up to nine viewpoints based on uncovered area gain and travel cost, stopping when the target coverage is reached or no useful candidate remains. An example viewpoint plan for the Workshop Lab is shown in Figure 3.

b) Search, Detection, and Approach: The robot visits the selected viewpoints using on-demand frame capture. At each reached viewpoint, it performs a small in-place yaw sweep of three snapshots over $\pm 10^\circ$, waiting for the detector to run inference on each frame before continuing. Up to three additional snapshots are requested while traveling to each viewpoint, subject to minimum time and distance between the two viewpoints. Detection evidence accumulates across distinct observations, and the target is confirmed after detections in at least two observations, which can

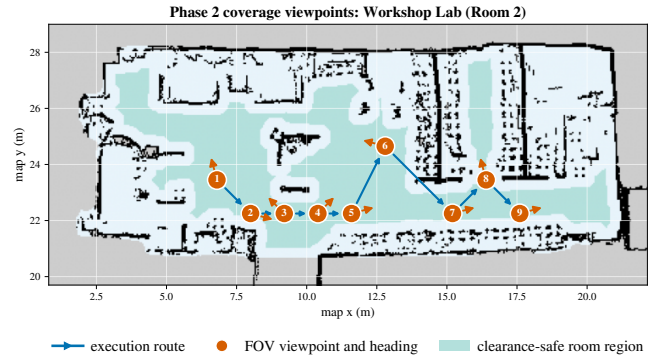


Fig. 3. Phase 2 viewpoint plan for the Workshop Lab (Room 2).

come from sweep angles or a mixture of stationary and in-motion observations. Upon confirmation, the robot switches to a stop-look-move controller: each cycle acquires a fresh snapshot and then fires either a yaw correction ($\leq 12^\circ$) or a short forward burst (0.35 m at 0.15 m/s). If the target is lost, the robot performs a stationary reacquisition sweep instead of advancing to a stale detection. The approach terminates when the height of the target bounding box reaches at least 0.50 of the frame height.

VII. EXPERIMENTAL METHODOLOGY

A. Real-World Evaluation

Real-world experiments were conducted in the six-room academic building shown in Figure 4. We evaluated the following four scenarios.

- 1) **One-robot Phase 1** (10 runs): A single robot classifies all six rooms sequentially, establishing the single-observer baseline.
- 2) **Three-robot Phase 1, single-visit** (10 runs): Each room receives one visit, a second visit is triggered if the committed confidence falls below 0.40, and a third visit is triggered on label disagreement. This measures the benefit of adaptive re-observation.
- 3) **Three-robot Phase 1, two-visit mandatory** (10 runs): Every room is visited by at least two distinct robots regardless of confidence, and a third robot is sent on label disagreement. This is the primary coordination protocol of the paper.
- 4) **Three-robot Phase 2, hint-based routing** (15 runs): After Phase 1, open-vocabulary queries are issued (a blue backpack or an umbrella) with room hints of varying specificity (a single room (Workshop Lab), two rooms (the office areas), or all four laboratories), activating one, two, or three robots respectively. These runs validate that hint-based routing scales the search team to the hinted room set and that coverage search and the stop-look-move approach execute correctly on hardware.

B. Controlled Simulation Evaluation

Since collecting many repeated trials with physical robots is time-consuming and introduces uncontrolled effects such as battery states, wireless congestion, and manual recoveries, the hardware evaluation is complemented with a controlled Gazebo simulation study that preserves the full algorithmic pipeline. The simulated world contains six apartment room

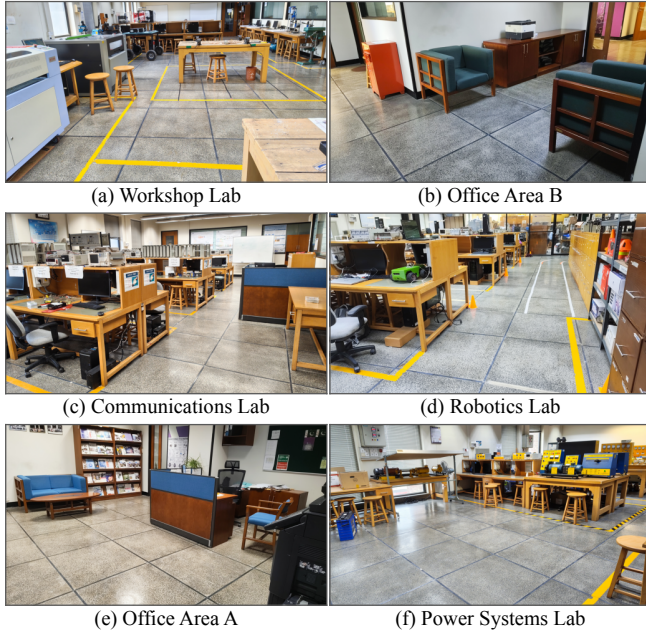


Fig. 4. Representative views of the six-room target environment used in the real-world evaluation. The rooms share common visual cues, motivating the need for multi-view evidence accumulation and belief fusion for room classification.

classes: kitchen, office, bedroom, bathroom, living room, and conference room. These classes and their furnishings are intentionally simulation specific and are not treated as equivalent to the six laboratory and office area classes present in the real-world environment. The simulation therefore evaluates the repeatability and coordination benefits rather than how well the pipeline conforms to a specific real-world environment.

Fifty Phase 1 runs are collected for each condition, yielding 300 settled room decisions per condition. Ground-truth room identities are used only for evaluation and are never provided to the navigation or fusion modules. For Phase 2, a controlled memory ablation is performed and evaluated using 10 two-object missions per condition. For these conditions, two factors are varied: team size (one or three robots) and semantic memory (enabled or disabled). With memory enabled, robots use the shared semantic map to navigate directly to the relevant room. With memory disabled, each robot independently classifies rooms using the VLM before searching, and these private classifications are not written or fused to the shared map. Disabled memory runs, therefore, measure the cost of decentralized room discovery.

All Phase-2 conditions used Qwen3-VL-4B for private room discovery, YOLOE for open-vocabulary detection, and the same coverage viewpoint planner, stop-look-move controller, and continuous simulated camera streams. Phase 2 runs started from a bootstrapped semantic map, so the reported Phase 2 times exclude Phase 1.

C. Metrics

To evaluate classification performance, we report room-level accuracy across all settled decisions, alongside Macro-F1, negative log-likelihood (NLL), Brier score [32], and ten-bin top-label expected calibration error (ECE) [33], [34]. Because evidence-table scores yield normalized beliefs rather than strictly calibrated posterior probabilities, these

metrics serve as relative diagnostics of belief quality. Phase 1 efficiency is measured using makespan and total team path length.

For Phase 2, we report complete two-object mission success, per-object confirmed-detection success, Phase 2 makespan T_{P2} , query-to-confirmed-detection time, total team path length L_{team} , and cumulative private room-discovery work T_{disc} .

VIII. RESULTS

A. Phase-1 Results

Table I reports results for three real-world Phase 1 conditions: one robot with one visit per room, three robots with one mandatory visit and additional visits triggered by low confidence or label disagreement, and three robots with two mandatory visits plus a disagreement-triggered third visit.

TABLE I
PHASE-1 REAL-WORLD RESULTS OVER 10 STABLE RUNS PER CONDITION.

Metric	One robot	Three robots (1 per room)	Three robots (2 per room)
Fused room accuracy (%)	88.33	91.67	98.33
Misclassified visits	7/60	8/70	9/127
Macro-F1	0.8445	0.9022	0.9311
NLL ↓	0.6140	0.5882	0.5443
Brier score ↓	0.2690	0.2574	0.2325
ECE ↓	0.3502	0.3303	0.3054
Makespan (s) – Initial visits	926.91	295.65	721.30
– All revisits	-	470.40	789.42
Team path (m) – Initial visits	198.61	141.94	334.78
– All revisits	-	222.19	393.93

A monotonic improvement in classification accuracy and the chosen scoring metrics across these experimental configurations is observed. Compared with the single-robot, single-visit baseline accuracy of 88.33%, the three-robot confidence-triggered revisit strategy increases accuracy to 91.67%, a gain of 3.34 percentage points. Requiring two visits per room raises accuracy to 98.33%, 10.00 percentage points above the baseline, corresponding to only one misclassification among 60 room classifications across 10 runs.

The multi-robot advantage is greatest when each room requires only one visit. The three-robot team completes the initial visits in 295.65 s, providing a $3.13\times$ speedup over the single-robot baseline of 926.91 s. Spatial parallelism also reduces the team path length from 198.61 m to 141.94 m, a 28.5% reduction. When revisits are included, the makespan and team path increase to 470.40 s and 222.19 m, respectively, reducing the speedup to $1.97\times$ and raising travel by 11.9% relative to the single-robot baseline. This diminished advantage reflects the additional travel required to revisit rooms outside the robots' initial assignments.

Beyond basic accuracy, this positive trend is consistently reflected in other key probabilistic and classification metrics. Specifically, as the visitation frequency is increased from one robot to the mandatory two-visit multi-robot setup, the Macro-F1 score improves from 0.8445 to 0.9311. Furthermore, uncertainty and calibration errors drop significantly: NLL decreases from 0.6140 to 0.5443, the Brier score drops from 0.2690 to 0.2325, and ECE reduces from 0.3502 to 0.3054.

The performance gain came at the cost of additional time and travel. The confidence-triggered three-robot condition was the most efficient overall, achieving both a lower

makespan and a shorter team path than the mandatory two-visit condition. The single-robot baseline travelled less than the mandatory two-visit condition, but because it visited all rooms sequentially, it required a longer makespan. Relative to confidence-triggered verification, mandatory revisits increased makespan and total team travel by 68% and 77%, respectively, demonstrating a clear accuracy-efficiency trade-off.

TABLE II
PHASE-1 SIMULATION RESULTS OVER 50 STABLE RUNS PER CONDITION.

Metric	One robot	Three robots
Room accuracy (%)	69.7	88.7
Misclassified decisions	91/300	34/300
Macro-F1	0.705	0.885
NLL ↓	1.256	1.189
Brier score ↓	0.605	0.572
ECE ↓	0.367	0.559
Makespan (s)	332.6 ± 8.5	536.1 ± 66.4
Team path (m)	63.0 ± 0.3	158.5 ± 12.8

Simulation exhibited a similar trend. Across 50 trials per condition, corresponding to 300 room decisions per condition, the three-robot fusion strategy increased room-classification accuracy from 69.7% to 88.7% and reduced errors from 91 to 34, a 62.6% reduction. Of the 300 three-robot decisions, 110 required a third tie-breaking visit, after which 82 (74.5%) were classified correctly. These results demonstrate that conflict-triggered verification suppresses many single-view errors. The remaining errors are concentrated among visually similar room pairs, indicating correlated ambiguity across viewpoints rather than independent observational noise.

Although the NLL and Brier score improved with three robots, ECE increased from 0.367 to 0.559. Fusion considerably improves correctness without increasing the normalized top-label belief, so the fused system is under-confident rather than artificially overconfident: in the most populated 0.2-0.3 belief bin, accuracy increased from 56.9% for one robot to 81.4% for three robots.

B. Phase-2 Results

We evaluated the three hint-routing conditions defined in Section VII through five hardware trials per condition (15 trials total), using a previously constructed shared semantic map. All 15 Phase 2 search-and-approach missions succeeded using the stored map. The routing mechanism activated robots according to hint breadth: one robot searched the single-room Workshop Lab condition, two robots divided the two-office condition, and three robots divided the four-laboratory condition. These trials validate coverage-based search and stop-look-move execution on physical hardware across all three team sizes. Because the conditions involved different candidate-room sets and the three-robot condition also used a different target object, their timing and path metrics do not constitute a controlled comparison of team size. That comparison was instead performed under matched conditions in the simulation memory ablation reported in Table III.

The widening room hint, from the single Workshop Lab to the two office areas to all four laboratories, activates

one, two, and three robots respectively on the nearest-claim assignment. Query-to-confirmed-detection time and team path length increased accordingly, from 201.9 s and 23.2 m with one robot in one room to 282.3 s and 102.1 m with three robots dividing four rooms (one robot searches two sequentially). The gap between Phase 2 makespan T_{P2} and query-to-confirmed-detection time stayed stable across conditions (32.2 s, 35.6 s, 41.1 s). Because only a single robot executes the physical approach once localized, scaling the team accelerates search coverage rather than approach time.

TABLE III
PHASE-2 SIMULATION MEMORY ABLATION OVER 10 SEQUENTIAL TWO-OBJECT MISSIONS PER CONDITION.

Condition	T_{P2} (s)	L_{team} (m)	T_{disc} (s)
1R, memory	112.3 ± 4.0	29.7 ± 0.7	–
1R, no memory	310.5 ± 13.9	63.0 ± 4.5	200.0 ± 4.3
3R, memory	90.3 ± 5.9	20.1 ± 0.9	–
3R, no memory	307.6 ± 9.9	104.2 ± 0.9	614.2 ± 18.6

Table III presents the simulation memory ablation using three metrics: Phase 2 makespan T_{P2} , measured from query dispatch to target arrival; total team path length L_{team} ; and cumulative private room-discovery time T_{disc} , incurred when robots independently classify rooms without access to the shared semantic map. Relative to the memory-enabled configurations, disabling semantic memory increased T_{P2} by 176.5% and 240.6% for the one- and three-robot teams, respectively. It also increased L_{team} by 112.1% and 418.4%, respectively. With memory enabled, the three-robot team reduced makespan by 19.6% and team travel by 32.3% relative to the single-robot configuration.

The no-memory ablation shows that team size alone provides negligible efficiency gains. Without shared semantic memory, three robots reduced makespan by only 0.9% relative to one robot, while performing $3.1\times$ more private discovery work (614.2 s versus 200.0 s) and traveling 65.4% farther.

C. VLM Ablation Study

The VLM used in this work is evaluated by comparing 11 open-source VLMs ranging from 1.5 to 4 billion parameters on room classification, holding the prompt description, evidence table, and substring matching fixed for a fair comparison. The evaluation used a curated dataset composed of 10 sets of frames for each of the 6 rooms, with each set containing 7 to 9 clear, non-duplicate frames totaling to 494 real-world frames. Each VLM was run on this dataset ten times with sampling $T = 0.7$ to evaluate their performance.

Five metrics are reported: runtime, peak memory usage, mean room accuracy, mean room confidence, and mean frame reliability. Accuracy and confidence are calculated by averaging the number of correctly classified rooms and the top room beliefs respectively, while reliability measures the prediction consistency of a specific frame across the 10 trials.

The accuracy varied considerably among the 11 models. Four models stood out with over 90% accuracy: Qwen3.5-4B, Qwen3-VL-4B, InternVL3.5-4B, and LFM2.5-VL-3B. Qwen3.5-4B achieved the highest raw accuracy, but Qwen3-VL-4B was chosen for deployment. While it trailed by only 1.3% in accuracy, it showed higher

TABLE IV
COMPARISON OF 11 VLMS ON ROOM CLASSIFICATION. SORTED BY
ACCURACY, DESCENDING.

Model	Params	Runtime (s/frame)	Peak VRAM (GB)	Accuracy (%)	Conf.	Reliability (%)
Qwen3.5-4B	4B	5.65	3.68	95.8	0.52	74.0
Qwen3-VL-4B	4B	3.91	3.33	94.5	0.65	84.6
InternVL3.5-4B	4B	5.08	4.25	93.2	0.46	76.0
LFM2.5-VL-3B	3B	1.84	6.06	92.3	0.46	75.4
Ovis2.5-2B	2B	3.82	3.58	81.7	0.45	74.4
LFM2-VL-1.6B	1.6B	1.56	3.13	81.2	0.42	65.3
InternVL3-2B	2B	2.51	2.09	79.2	0.37	68.7
Qwen2.5-VL-3B	3B	3.37	2.72	76.0	0.39	68.7
Phi-3.5-Vision	4.2B	7.16	5.28	61.8	0.27	61.1
SmolVLM2-2.2B	2.2B	2.12	2.56	59.5	0.25	68.4
FastVLM-1.5B	1.5B	2.49	1.51	41.8	0.20	81.9

per-room confidence and per-frame reliability and ran 1.74s faster using ~350 MB less VRAM on average. Note that the VLM benchmark used a curated subset of clear, discriminative frames and therefore measured per-frame model performance under favorable visibility rather than end-to-end room-classification accuracy. Therefore, its results are not directly comparable with the mission-level results in Table I.

Although the difference between 95% and 90% accuracy seems modest in these experiments, this difference has a very large implication in the efficiency of the experiment. Since the robots must classify a room at least 12 times over the course of the experiment, the VLM accuracy sets the chance of a label conflict that triggers a costly third-robot tie-break visit. With per-room agreement probability $p = (\text{accuracy})^2$, the chance that all six rooms settle on the first sweep is p^6 . Qwen3.5-4B (95.8%) and LFM2.5-VL-3B (92.3%) both exceed 90% accuracy, yet this gap separates a 59.76% from a 38.23% chance of a clean first sweep, a 21.53-point difference. Additionally, for this experiment, the number of parameters of the model do not correlate with the accuracy, as LFM2-VL-1.6B with 1.6 billion parameters outperformed both InternVL3-2B and Qwen2.5-VL-3B while also classifying each frame at very high speeds (<1.6s on average).

Peak GPU memory footprint also proved to be a poor predictor of downstream performance. InternVL3-2B required only 2.09GB and 2.51s per frame compared to Phi-3.5-Vision’s 5.28GB and 7.16s, yet the smaller and faster InternVL3-2B model achieved a 17.4-point higher accuracy. Among the strongest performers, Qwen3.5-4B achieved the highest accuracy at 95.8%, while using 570 MB less VRAM than InternVL3.5-4B which achieved 93.2%. Qwen3-VL-4B came within 1.3 points of the best accuracy while using only 3.33 GB of VRAM.

As an external baseline, we evaluated exemplar-conditioned Qwen3-VL-4B classification [35] using 1–6 labeled reference images per room, disjoint from the test set. On identical frames, it achieved 68.5% accuracy at $N = 3$ and 62.4–74.8% across all N , without monotonic improvement, compared with 94.5% for Hive-Nav. It also performed worse than a generic-text-hint zero-shot baseline (80.0%) even with the added visual grounding. Despite its lower accuracy, its mean confidence exceeded Hive-Nav’s (0.83 versus 0.65). Errors were dominated by directional confusions: the Communications Lab was misclassified 93% of the time, with high-confidence rationales treating generic furniture as class-specific. Thus, visual exemplars alone did not resolve hallucination-driven overconfidence.

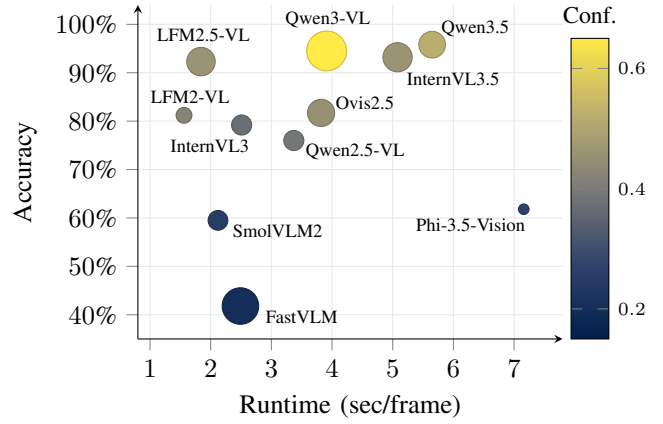


Fig. 5. VLM room-classification accuracy vs. runtime for 11 models. Marker size encodes frame-level reliability; color encodes confidence.

IX. DISCUSSION

A. The Role of Shared Semantic Memory in Multi-Robot Efficiency

Multi-robot deployment pays off only through the shared semantic map, since Hive-Nav serializes all VLM inference through one central GPU. Once the Phase 1 belief map exists, the team routes directly to the target room and the shared GPU is dedicated almost entirely to Phase 2 detection. Without shared memory, every robot must independently run Phase 1 classification, and because these requests queue behind the same lock, three robots do not triple classification throughput, they triple queue contention instead. The results in Table III establish shared semantic memory, not robot count alone, as the principal enabler of Hive-Nav’s Phase 2 efficiency gains.

B. Impact of Shared Wi-Fi Network

All inference traffic traversed the university’s uncontrolled shared Wi-Fi network. During Phase 1 and Phase 2 trials, 1-Hz heartbeat probes recorded median round-trip times below 10 ms but highly variable tails. Per-run packet loss ranged from 1.1% to 20.8%, while 95th-percentile latency ranged from hundreds of milliseconds to several seconds. These metrics showed no monotonic relationship with team size, indicating substantial ambient network variability and supporting the on-demand frame-capture design described in Section IV.

C. Limitations and Future Work

Our physical evaluation focused on a three-robot team in a six-room, single-floor environment, comprising 10 Phase 1 trials and five Phase 2 trials per routing condition (compared to 50 and 10 runs per condition in simulation). This physical sample size reflects the significant operational overhead of multi-robot field trials, including post-trial robot retrieval, state resets, and battery cycling. Nevertheless, by demonstrating the coordination and fusion pipeline across both controlled simulations and physical hardware, this work establishes a validated empirical foundation for scaling to larger fleets, heterogeneous teams, multi-floor topologies, and broader semantic categories.

The evidence table was manually designed for the six room classes, and its signed weights represent ranked evidence rather than calibrated likelihoods. Consequently, trans-

ferring Hive-Nav to a different room taxonomy or substantially different environment may require redesigning the phrase-to-class mappings. Automatically mining phrases or learning their weights could improve transferability, while hallucination-mitigation methods such as attention tracing and origin-point regeneration [36] could improve the reliability of VLM inventories.

Due to the constrained onboard compute of our Raspberry-Pi platforms, VLM inference was serialized on a single shared GPU. While this centralized architecture supported our three-robot team reliably over ambient Wi-Fi, scaling to larger fleets remains unverified; in larger deployments, GPU queueing and scheduling delays are likely to surpass network variability as the dominant system bottleneck.

X. CONCLUSION

This paper presented Hive-Nav, a two-phase classify-then-search framework for open-vocabulary multi-robot target search, deployed on three ground robots using centralized inference over shared campus Wi-Fi. Multi-visit belief fusion increased real-world room-classification accuracy from 88.3% to 98.3%, and all 15 physical Phase 2 search-and-approach trials succeeded using the stored map. Controlled simulations clarified the value of reusing Phase 1 beliefs. With shared memory, three robots reduced makespan by 19.6% and total travel by 32.3% relative to one robot under the same condition. Without stored beliefs, three robots were only 0.9% faster despite performing $3.1\times$ the room-discovery work. These results demonstrate that shared semantic memory, rather than team size alone, is essential for efficient multi-robot coordination under constrained inference and communication, providing an empirical baseline for future exploration in decentralized multi-floor deployments.

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